# Package 'enviro'

October 25, 2026

**Title** Displays data of pimoroni enviro from prometheus

**Version** 0.0.0.9000

**Description** A Shiny application, built as a golem package, that displays the readings of Pimoroni Enviro and Enviro+ sensors exported to Prometheus: gauges with the latest readings, and a time series chart per sensor.

**License** GPL-3

**URL** https://sedzinfo.github.io/enviro/, https://github.com/sedzinfo/enviro

**BugReports** https://github.com/sedzinfo/enviro/issues

**Encoding** UTF-8

**Roxygen** list(markdown = TRUE)

**Imports** config,
golem,
htmltools,
httr,
jsonlite,
plotly,
shiny,
shinydashboard,
stats,
zoo

**Suggests** testthat (>= 3.0.0)

**Config/testthat/edition** 3

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**Config/roxygen2/version** 8.1.1

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`get_data` *Get Data from Prometheus*

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## Description

Downloads the readings of all the sensors of a Pimoroni Enviro or Enviro+ from Prometheus, at about 1000 points in time between `start_time` and `end_time`.

## Usage

```
get_data(
  start_time = format(Sys.time() - 60 * 60 * 24 * 7, "%Y-%m-%dT%H:%M:%SZ"),
  end_time = format(Sys.time(), "%Y-%m-%dT%H:%M:%SZ"),
  urlq
)
```

## Arguments

| | |
|---|---|
| `start_time` | start time in this format: "2025-01-31T11:48:08Z" |
| `end_time` | end time in this format: "2025-02-07T11:48:33Z" |
| `urlq` | Prometheus url |

## Value

A data frame with the Date and the Timestamp of the readings and one column per sensor, or `NULL`, with a message, when the server cannot be reached or sends no data. Sensors whose readings never change are left out, as the device does not have them.

## Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
start_time <- format(Sys.time() - 60 * 60 * 24 * 7, "%Y-%m-%dT%H:%M:%SZ")
end_time <- format(Sys.time(), "%Y-%m-%dT%H:%M:%SZ")
filtered_df <- get_data(start_time = start_time, end_time = end_time, urlq = prometheus_url[2])
head(filtered_df)
filtered_df <- get_data(start_time = start_time, end_time = end_time, urlq = prometheus_url[1])
head(filtered_df)
```

pimoroni *Run the Pimoroni Enviro app*

## Description

Starts a Shiny app with the readings of Pimoroni Enviro and Enviro+ sensors exported to Prometheus: gauges with the latest readings in the Current tab, and a chart per sensor in the Time Series tab.

## Usage

```
pimoroni(
  prometheus_url,
  timezone = "Europe/Bucharest",
  onStart = NULL,
  options = list(),
  enableBookmarking = NULL,
  uiPattern = "/",
  ...
)
```

## Arguments

| | |
|---|---|
| prometheus_url | Character vector of Prometheus query_range urls, one per device, such as "http://pip1.crabdance.com:150 |
| timezone | The time zone the readings are shown in, while the app runs. |
| onStart | See shiny::shinyApp(). |
| options | See shiny::shinyApp(). |
| enableBookmarking | See shiny::shinyApp(). |
| uiPattern | See shiny::shinyApp(). |
| ... | Further options passed to golem::with_golem_options(). |

## Value

A shiny::shinyApp() object. Printing it, or calling pimoroni() at the console, opens the app in the browser.

## Examples

```
if (interactive()) {
  pimoroni(prometheus_url = c(
    "http://pip1.crabdance.com:1507/api/v1/query_range",
    "http://pip1.crabdance.com:1505/api/v1/query_range"
  ))
}
```

## plot_gauge *Plot Gauge*

### Description

Plots the last reading of one sensor as a gauge, with the ideal range in green and a short note on what the gas and particle sensors measure.

### Usage

```
plot_gauge(data, variable = "Temperature")
```

### Arguments

| | |
|---|---|
| data | data frame from get_data() |
| variable | one of "Temperature" "Humidity" "Pressure" "Lux" "Proximity" "NH3" "Reducing" "Oxidising" "PM1" "PM2.5" "PM10" |

### Value

A plotly object.

### Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
filtered_df <- get_data(urlq = prometheus_url[1])
plot_gauge(data = filtered_df, variable = "Temperature")
plot_gauge(data = filtered_df, variable = "Humidity")
plot_gauge(data = filtered_df, variable = "PM2.5")
```

## plot_time_series *Plot Time Series*

### Description

Plots the readings of one sensor, with their moving average, their mean, and a summary of the last reading, mean, minimum, maximum, standard deviation, range and dates on the right.

### Usage

```
plot_time_series(data, variable = "Temperature", scale = "", k = 30)
```

## Arguments

| | |
|---|---|
| data | data frame from get_data() |
| variable | one of "Temperature" "Humidity" "Pressure" "Lux" "Proximity" "NH3" "Reducing" "Oxidising" "PM1" "PM2.5" "PM10" |
| scale | title of the y axis |
| k | integer width of the rolling window. Must be odd for rollmedian |

## Value

A plotly object.

## Examples

```
prometheus_url <- c(
  "http://pip1.crabdance.com:1507/api/v1/query_range",
  "http://pip1.crabdance.com:1505/api/v1/query_range"
)
filtered_df <- get_data(urlq = prometheus_url[1])
plot_time_series(data = filtered_df, variable = "Temperature")
plot_time_series(data = filtered_df, variable = "Humidity")
plot_time_series(data = filtered_df, variable = "Pressure")
plot_time_series(data = filtered_df, variable = "PM2.5")
```

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